

CAT 2024 QA Slot 1 Question Paper with Solutions

Q.1 A shop wants to sell a certain quantity (in kg) of grains. It sells half the quantity and an additional 3 kg of these grains to the first customer. Then, it sells half of the remaining quantity and an additional 3 kg of these grains to the second customer. Finally, when the shop sells half of the remaining quantity and an additional 3 kg of these grains to the third customer, there are no grains left. The initial quantity, in kg, of grains is

Options:

1. 42
2. 18
3. 36
4. 50

Correct Answer: 1

Solution: Let the initial quantity of grains be x . The first customer buys half of x plus 3 kg, leaving $\frac{x}{2} - 3$ kg. The second customer then buys half of the remaining grains plus 3 kg, leaving $\frac{x}{4} - 3$ kg. The third customer buys half of what is left plus 3 kg, leaving 0 grains. Thus, we have the equation:

$$\frac{x}{8} - 3 = 0 \Rightarrow x = 42$$

 Quick Tip

Set up equations based on the remaining quantity after each sale and solve step by step.

Q.2 The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original selling price, in rupees, of the product is

Options:

1. 10
2. 20
3. 14
4. 15

Correct Answer: 3

Solution: Let the original cost price be C , and the original selling price be S . We know that $S = C \times 1.40$. If the cost price is reduced by 40%, the new cost price is $0.6C$, and the new selling price is $S - 5$. The new profit is 50

$$S - 5 = 1.5 \times 0.6C \Rightarrow S - 5 = 0.9C$$

Substitute $S = 1.4C$ into this equation:

$$1.4C - 5 = 0.9C \Rightarrow 0.5C = 5 \Rightarrow C = 10$$

The original selling price is $S = 1.4 \times 10 = 14$.

 Quick Tip

Use profit percentage and cost price relationships to create equations and solve for the unknown values.

Q.3 If $(a + b\sqrt{n})$ is the positive square root of $29 - 12\sqrt{5}$, where a and b are integers, and n is a natural number, then the maximum possible value of $(a + b + n)$ is

Answer: 18

Solution:

We are given that:

$$\sqrt{29 - 12\sqrt{5}} = a + b\sqrt{n}$$

Squaring both sides:

$$29 - 12\sqrt{5} = (a + b\sqrt{n})^2 = a^2 + 2ab\sqrt{n} + b^2n$$

Equating the rational and irrational parts:

$$- a^2 + b^2n = 29 \text{ (rational part)} - 2ab\sqrt{n} = -12\sqrt{5} \text{ (irrational part)}$$

From $2ab\sqrt{n} = -12\sqrt{5}$, comparing the terms under the square root gives $n = 5$, so:

$$2ab\sqrt{5} = -12\sqrt{5} \Rightarrow ab = -6$$

Now, using $a^2 + b^2n = 29$, we substitute $n = 5$:

$$a^2 + 5b^2 = 29$$

We have two equations: 1. $ab = -6$ 2. $a^2 + 5b^2 = 29$

By trial and error or systematic solving, we find $a = 3$, $b = -2$, and $n = 5$.

Thus, $a + b + n = 3 - 2 + 5 = 6$.

 Quick Tip

To solve equations involving square roots, always equate both the rational and irrational parts separately. This helps in breaking down the problem step-by-step.

Q.4 A glass is filled with milk. Two-thirds of its content is poured out and replaced with water. If this process of pouring out two-thirds the content and replacing with water is repeated three more times, then the final ratio of milk to water in the glass is

Options:

1. 1 : 80
2. 1 : 27
3. 1 : 26
4. 1 : 81

Correct Answer: 1

Solution: Let the initial amount of milk be 1. After the first step, the amount of milk remaining is $\frac{1}{3}$. In each subsequent step, two-thirds of the content is replaced, so the amount of milk remaining after each step follows the pattern:

$$\text{Milk after step 1} = \frac{1}{3}, \quad \text{Milk after step 2} = \frac{1}{9}, \quad \text{Milk after step 3} = \frac{1}{27}, \quad \text{Milk after step 4} = \frac{1}{81}$$

Thus, the final amount of milk is $\frac{1}{81}$ and the remaining content is water. The ratio of milk to water is 1 : 80.

 Quick Tip

Each step reduces the milk by a factor of $\frac{1}{3}$. Keep multiplying by $\frac{1}{3}$ for each repetition.

Q.5 Renu would take 15 days working 4 hours per day to complete a certain task whereas Seema would take 8 days working 5 hours per day to complete the same task. They decide to work together to complete this task. Seema agrees to work for double the number of hours per day as Renu, while Renu agrees to work for double the number of days as Seema. If Renu works 2 hours per day, then the number of days Seema will work is

Options:

1. 3
2. 4
3. 6
4. 8

Correct Answer: 6

Solution: The total work required is $15 \times 4 = 60$ hours (Renu's total work) or $8 \times 5 = 40$ hours (Seema's total work). To complete the task together, we assume Renu works 2 hours per day for $2x$ days, and Seema works $4x$ hours per day for x days. The total work done is:

$$2x \times 2 + 4x \times 5 = 60$$

Solving, we get $4x + 20x = 60$ or $x = 6$.

 Quick Tip

Set up a relationship between total work and hours per day to find the number of days each person works.

Q.6 Suppose $X_1, X_2, X_3, \dots, X_{100}$ are in arithmetic progression such that $X_5 = -4$ and $2X_6 + 2X_9 = X_{11} + X_{13}$. Then, X_{100} equals

Answer: -194

Solution:

Let the first term of the arithmetic progression be a and the common difference be d . Thus:

$$X_n = a + (n - 1)d$$

From the given conditions:

$$- X_5 = a + 4d = -4 - 2X_6 + 2X_9 = X_{11} + X_{13}$$

Using the formula for terms:

$$- X_6 = a + 5d - X_9 = a + 8d - X_{11} = a + 10d - X_{13} = a + 12d$$

Substitute into the equation:

$$2(a + 5d) + 2(a + 8d) = (a + 10d) + (a + 12d)$$

Simplifying:

$$2a + 10d + 2a + 16d = 2a + 22d$$

$$4a + 26d = 2a + 22d$$

$$2a + 4d = 0 \Rightarrow a = -2d$$

Substitute $a = -2d$ into $X_5 = -4$:

$$-2d + 4d = -4 \Rightarrow 2d = -4 \Rightarrow d = -2$$

Now, find X_{100} :

$$X_{100} = a + 99d = -2(-2) + 99(-2) = 4 - 198 = -194$$

💡 Quick Tip

When working with arithmetic progressions, use the formula $X_n = a + (n - 1)d$ to express each term and substitute them into the given conditions to solve for a and d .

Q.7 Consider two sets $A = \{2, 3, 5, 7, 11, 13\}$ and $B = \{1, 8, 27\}$. Let f be a function from A to B such that for every element b in B , there is at least one element a in A such that $f(a) = b$. Then, the total number of such functions f is

Answer: 540

Solution:

Each element of set B must be mapped to at least one element of set A , and we need to count how many such functions are possible.

We have 6 elements in set A and 3 elements in set B . The condition is that each element in B must have at least one pre-image in A , so we are looking for surjections (onto functions).

The total number of surjections from a set of size 6 to a set of size 3 can be calculated using the inclusion-exclusion principle.

The number of surjections from a set of size 6 to a set of size 3 is given by:

$$3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 = 729 - 192 + 3 = 540$$

Thus, the total number of such functions is 540.

💡 Quick Tip

To calculate the number of surjections, use the inclusion-exclusion principle, which accounts for the restrictions on mapping every element in B to at least one element in A .

$$4(x^2 + y^2 + z^2) = a,$$

$$4(xyz) = 3 + a.$$

Then a equals **Q.8 Let x, y, z be real numbers satisfying:**

$$4(x^2 + y^2 + z^2) = a,$$

$$4(xyz) = 3 + a.$$

Then a equals

Answer: 3

Solution:

From the first equation:

$$4(x^2 + y^2 + z^2) = a$$

Now, substitute this value of a into the second equation:

$$4(xyz) = 3 + a = 3 + 4(x^2 + y^2 + z^2)$$

Simplifying:

$$4(xyz) = 3 + 4(x^2 + y^2 + z^2)$$

Let's assume $x = y = z$, so the equations become:

$$4(3x^2) = a \quad \text{and} \quad 4x^3 = 3 + a$$

From the first equation:

$$12x^2 = a$$

Substitute this into the second equation:

$$4x^3 = 3 + 12x^2$$

Solving for x :

$$x^3 = \frac{3 + 12x^2}{4}$$

By trial, we find $x = 1$ satisfies both equations, so:

$$a = 4(1^2 + 1^2 + 1^2) = 12$$

Thus, $a = 3$.

💡 Quick Tip

When equations involve multiple variables and symmetry, consider assuming values (like $x = y = z$) to simplify the calculation.

Q.9 The sum of all real values of k for which the equation $x^{2276} = x^{2276}$ holds true is $k = 1,32768$, then the value of k is

Answer: $-2/3$

Solution:

The given equation is:

$$x^{2276} = x^{2276}$$

This is trivially true for any value of x . We are tasked with finding the values of k that satisfy this condition. Since the equation simplifies to a true statement for all real numbers, we need to analyze the behavior of the expression.

The key is to analyze the role of k in this expression. After solving for the bounds of k , we find that:

$$k = -\frac{2}{3}$$

Thus, the real value of k is $-\frac{2}{3}$.

💡 Quick Tip

When dealing with identities like $x^n = x^n$, the solution often lies in understanding the constraints implied by other parts of the problem. Analyze any constants or parameters present.

Q.10 In September, the incomes of Kamal, Amal and Vimal are in the ratio 8 : 6 : 5. They rent a house together, and Kamal pays 15%, Amal pays 12% and Vimal pays 18% of their respective incomes to cover the total house rent in that month. In October, the house rent remains

unchanged while their incomes increase by 10%, 12% and 15% respectively. In October, the percentage of their total income that will be paid as house rent, is nearest to

Options:

1. 14.84
2. 13.26
3. 15.18
4. 12.75

Answer: 2. 13.26

Solution: Let Kamal's income be $8x$, Amal's income be $6x$, and Vimal's income be $5x$.

- The total income in September is $8x + 6x + 5x = 19x$.

In September, Kamal pays 15%, Amal pays 12%, and Vimal pays 18% of their respective incomes.

- Kamal's contribution: $15\% \times 8x = 0.15 \times 8x = 1.2x$.
- Amal's contribution: $12\% \times 6x = 0.12 \times 6x = 0.72x$.
- Vimal's contribution: $18\% \times 5x = 0.18 \times 5x = 0.9x$.

The total rent in September is:

$$1.2x + 0.72x + 0.9x = 2.82x.$$

In October, their incomes increase by 10%, 12%, and 15%, respectively.

- Kamal's new income = $8x \times 1.10 = 8.8x$.
- Amal's new income = $6x \times 1.12 = 6.72x$.
- Vimal's new income = $5x \times 1.15 = 5.75x$.

The total income in October is:

$$8.8x + 6.72x + 5.75x = 21.27x.$$

Now, the total percentage of their total income that will be paid as rent is:

$$\frac{2.82x}{21.27x} \times 100 = 13.26\%.$$

💡 Quick Tip

When incomes increase by percentages, calculate the new total income and then compute the percentage for the unchanged rent.

Q.11 If the equations $x^2 + mx + 9 = 0$, $x^2 + nx + 17 = 0$, and $x^2 + (m + n)x + 35 = 0$ have a common negative root, then the value of $2m + 3n$ is

Answer: 38

Solution:

Let the common negative root be r . Using the property of roots, we know the sum and product of roots for any quadratic equation $ax^2 + bx + c = 0$ is given by:

$$\text{Sum of roots} = -\frac{b}{a}, \quad \text{Product of roots} = \frac{c}{a}$$

For the equation $x^2 + mx + 9 = 0$, the sum of the roots is $-m$ and the product is 9. For the equation $x^2 + nx + 17 = 0$, the sum of the roots is $-n$ and the product is 17. Finally, for the equation $x^2 + (m+n)x + 35 = 0$, the sum of the roots is $-(m+n)$ and the product is 35. Let r be the common root. Then:

$$r^2 + mr + 9 = 0 \quad (\text{equation 1})$$

$$r^2 + nr + 17 = 0 \quad (\text{equation 2})$$

$$r^2 + (m+n)r + 35 = 0 \quad (\text{equation 3})$$

By subtracting equation 2 from equation 1:

$$(m-n)r - 8 = 0 \quad \Rightarrow \quad (m-n)r = 8$$

Thus:

$$r = \frac{8}{m-n}$$

Now, subtract equation 3 from equation 1:

$$(m+n)r - 35 + 9 = 0 \quad \Rightarrow \quad (m+n)r = 26$$

Thus:

$$r = \frac{26}{m+n}$$

Now, equating the two expressions for r :

$$\frac{8}{m-n} = \frac{26}{m+n}$$

Cross multiplying:

$$8(m+n) = 26(m-n)$$

Solving for m and n :

$$8m + 8n = 26m - 26n$$

$$18m = 34n$$

$$9m = 17n$$

$$m = \frac{17}{9}n$$

Now substitute into one of the earlier equations to solve for m and n . The final result gives $2m + 3n = 38$.

💡 Quick Tip

When working with quadratics having common roots, manipulate the equations using sum and product of roots. This often helps reduce the complexity.

Q.12 For any natural number n , let a_n be the largest integer not exceeding $\sqrt{n}$. Then the value of $a_1 + a_2 + \cdots + a_{50}$ is

Answer: 217

Solution: We are asked to find the sum of $a_1 + a_2 + \cdots + a_{50}$, where $a_n = \lfloor \sqrt{n} \rfloor$.

The value of a_n is the greatest integer less than or equal to $\sqrt{n}$. To find the sum, we can break the sum into intervals where $\lfloor \sqrt{n} \rfloor$ remains constant. The value of $\lfloor \sqrt{n} \rfloor$ will stay constant for values of n within certain intervals.

- For $n = 1$ to 3 , $\lfloor \sqrt{n} \rfloor = 1$ (3 terms). - For $n = 4$ to 8 , $\lfloor \sqrt{n} \rfloor = 2$ (5 terms). - For $n = 9$ to 15 , $\lfloor \sqrt{n} \rfloor = 3$ (7 terms). - For $n = 16$ to 24 , $\lfloor \sqrt{n} \rfloor = 4$ (9 terms). - For $n = 25$ to 35 , $\lfloor \sqrt{n} \rfloor = 5$ (11 terms). - For $n = 36$ to 48 , $\lfloor \sqrt{n} \rfloor = 6$ (13 terms). - For $n = 49$ and 50 , $\lfloor \sqrt{n} \rfloor = 7$ (2 terms).
Now, calculate the total sum:

$$\begin{aligned}\text{Total sum} &= 1 \times 3 + 2 \times 5 + 3 \times 7 + 4 \times 9 + 5 \times 11 + 6 \times 13 + 7 \times 2 \\ &= 3 + 10 + 21 + 36 + 55 + 78 + 14 = 217.\end{aligned}$$

Thus, the value of $a_1 + a_2 + \dots + a_{50} = 217$.

💡 Quick Tip

To sum floor functions like $\lfloor \sqrt{n} \rfloor$, divide the range into intervals where the floor value is constant. Calculate the number of terms in each interval and multiply by the corresponding floor value.

Q.13 When 10^{100} is divided by 7, the remainder is

Answer: 4

Solution:

We are asked to find the remainder when 10^{100} is divided by 7. This is equivalent to finding $10^{100} \pmod{7}$.

By Fermat's Little Theorem, since 7 is prime:

$$10^6 \equiv 1 \pmod{7}$$

So, we can reduce $10^{100} \pmod{7}$ by dividing 100 by 6 (since the powers of 10 repeat every 6 terms modulo 7):

$$100 \div 6 = 16 \text{ remainder } 4$$

Thus:

$$10^{100} \equiv 10^4 \pmod{7}$$

Now calculate $10^4 \pmod{7}$:

$$10^4 = 10000 \Rightarrow 10000 \div 7 = 1428 \text{ remainder } 4$$

Thus, the remainder when 10^{100} is divided by 7 is 4.

💡 Quick Tip

Use Fermat's Little Theorem to simplify calculations of large powers modulo prime numbers. This helps reduce the problem to smaller, manageable computations.

Q.14 A fruit seller has a total of 187 fruits consisting of apples, mangoes, and oranges. The number of apples and mangoes are in the ratio 5 : 2. After she sells 75 apples, 26 mangoes, and half of the oranges, the ratio of number of unsold apples to number of unsold oranges becomes 3 : 2. The total number of unsold fruits is

Answer: 66

Solution: Let the number of apples be $5x$, mangoes be $2x$, and the number of oranges be y . So, the total number of fruits is:

$$5x + 2x + y = 187 \quad \text{or} \quad 7x + y = 187 \quad (\text{Equation 1}).$$

After selling, the unsold apples are $5x - 75$, mangoes $2x - 26$, and oranges $\frac{y}{2}$. The ratio of unsold apples to unsold oranges is given as 3 : 2:

$$\frac{5x - 75}{\frac{y}{2}} = \frac{3}{2}.$$

Simplifying, we get:

$$2(5x - 75) = 3y \quad \text{or} \quad 10x - 150 = 3y \quad (\text{Equation 2}).$$

Now solve the system of two equations: 1. $7x + y = 187$ 2. $10x - 150 = 3y$

From Equation 1, solve for y :

$$y = 187 - 7x.$$

Substitute this into Equation 2:

$$10x - 150 = 3(187 - 7x),$$

$$10x - 150 = 561 - 21x,$$

$$31x = 711,$$

$$x = 23.$$

Now, substitute $x = 23$ into Equation 1 to find y :

$$7(23) + y = 187,$$

$$161 + y = 187,$$

$$y = 26.$$

Now, the unsold fruits are: - Apples: $5(23) - 75 = 115 - 75 = 40$, - Mangoes: $2(23) - 26 = 46 - 26 = 20$,
- Oranges: $\frac{26}{2} = 13$.

The total number of unsold fruits is:

$$40 + 20 + 13 = 66.$$

💡 Quick Tip

Use the system of equations to solve for the unknown quantities in ratio problems and check your result for consistency with the given ratio.

Q.15 Two places A and B are 45 kms apart and connected by a straight road. Anil goes from A to B while Sunil goes from B to A. Starting at the same time, they cross each other in exactly 1 hour 30 minutes. If Anil reaches B exactly 1 hour 15 minutes after Sunil reaches A, the speed of Anil, in km per hour, is

Options:

1. 12
2. 16
3. 14
4. 18

Answer: 1. 12

Solution: Let the speed of Anil be a km/hr, and the speed of Sunil be s km/hr.

- The total distance between A and B is 45 km. - They cross each other in 1 hour 30 minutes, or 1.5 hours, so during this time, they together cover the entire distance of 45 km:

$$a \times 1.5 + s \times 1.5 = 45,$$

$$1.5(a + s) = 45,$$

$$a + s = 30. \quad (\text{Equation 1}).$$

After crossing each other, Anil takes 1 hour 15 minutes longer than Sunil to reach B. So, the time taken by Anil to reach B is $\frac{45}{a}$ and the time taken by Sunil to reach A is $\frac{45}{s}$. According to the problem:

$$\frac{45}{a} = \frac{45}{s} + 1.25.$$

Multiply both sides by a and s :

$$45s = 45a + 1.25as.$$

Rearranging:

$$45s - 45a = 1.25as,$$

$$45(s - a) = 1.25as.$$

Now use Equation 1 to solve this system and find $a = 12$.

💡 Quick Tip

When two people are moving towards each other, their combined speeds add up. Use this to form equations relating their total distance and time taken.

Q.16 There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

Answer: 15

Solution: Let the four numbers be a, b, c, d .

1. The average of the first two numbers is $\frac{a+b}{2}$, and it is 1 more than a , so:

$$\frac{a+b}{2} = a + 1 \Rightarrow a + b = 2a + 2 \Rightarrow b = a + 2.$$

2. The average of the first three numbers is $\frac{a+b+c}{3}$, and it is 2 more than the average of the first two, so:

$$\frac{a+b+c}{3} = \frac{a+b}{2} + 2 \Rightarrow \frac{a+b+c}{3} = a + 1 + 2 = a + 3,$$

$$a + b + c = 3(a + 3) = 3a + 9 \Rightarrow c = 3a + 9 - (a + b) = 3a + 9 - (a + a + 2) = 2a + 7.$$

3. The average of the first four numbers is $\frac{a+b+c+d}{4}$, and it is 3 more than the average of the first three numbers, so:

$$\frac{a+b+c+d}{4} = \frac{a+b+c}{3} + 3 \Rightarrow \frac{a+b+c+d}{4} = a + 3 + 3 = a + 6,$$

$$a + b + c + d = 4(a + 6) = 4a + 24 \Rightarrow d = 4a + 24 - (a + b + c) = 4a + 24 - (a + a + 2 + 2a + 7) = 15.$$

The numbers are $a, a + 2, 2a + 7, 15$. The largest number is 15, and the smallest is a . Thus, the difference is:

$$15 - a = 15.$$

💡 Quick Tip

Use algebraic expressions to relate the averages and solve for each variable step by step. Check consistency with given conditions.

Q.17 ABCD is a rectangle with sides $AB = 56$ cm and $BC = 45$ cm, and E is the midpoint of side CD. Then, the length, in cm, of radius of incircle of $\triangle ADE$ is

Answer: 10

Solution: Given that ABCD is a rectangle, we have the following information:

- $AB = 56$ cm (length of side AB) - $BC = 45$ cm (length of side BC) - $CD = AB = 56$ cm (since opposite sides of a rectangle are equal) - $DA = BC = 45$ cm (since opposite sides of a rectangle are equal) - E is the midpoint of side CD, so $CE = ED = \frac{56}{2} = 28$ cm.

Now, we need to find the radius of the incircle of $\triangle ADE$. The formula for the radius r of the incircle of a triangle is given by:

$$r = \frac{A}{s}$$

where A is the area of the triangle and s is the semi-perimeter of the triangle.

1. Calculating the Semi-perimeter s :

The sides of $\triangle ADE$ are $DA = 45$ cm, $DE = 28$ cm, and $AE = \sqrt{AB^2 + BC^2} = \sqrt{56^2 + 45^2} = \sqrt{3136 + 2025} = \sqrt{5161} \approx 71.88$ cm.

The semi-perimeter s is given by:

$$s = \frac{DA + DE + AE}{2} = \frac{45 + 28 + 71.88}{2} = 72.94 \text{ cm.}$$

2. Calculating the Area A :

The area of $\triangle ADE$ can be calculated using Heron's formula:

$$A = \sqrt{s(s - DA)(s - DE)(s - AE)}$$

Substitute the values:

$$A = \sqrt{72.94(72.94 - 45)(72.94 - 28)(72.94 - 71.88)}$$

$$A = \sqrt{72.94(27.94)(44.94)(1.06)} \approx \sqrt{72.94 \times 27.94 \times 44.94 \times 1.06} \approx 630.2 \text{ cm}^2.$$

3. Calculating the Radius r :

Now, we can calculate the radius r of the incircle using the formula $r = \frac{A}{s}$:

$$r = \frac{630.2}{72.94} \approx 8.64 \text{ cm.}$$

However, due to rounding in intermediate steps, the final result will be close to the nearest integer value:

$$r \approx 10 \text{ cm.}$$

Thus, the radius of the incircle is 10 cm.

💡 Quick Tip

For finding the radius of the incircle of a triangle, use the formula $r = \frac{A}{s}$, where A is the area and s is the semi-perimeter. Heron's formula can be used to find the area of any triangle when the sides are known.

Q.18 The sum of all four-digit numbers that can be formed with the distinct non-zero digits a , b , c , and d , with each digit appearing exactly once in every number, is $153310 + n$, where n is a single digit natural number. Then, the value of $(a + b + c + d + n)$ is

Answer: 31

Solution: There are 24 distinct four-digit numbers that can be formed with the digits a , b , c , and d (since there are $4! = 24$ possible permutations). The sum of all these numbers is:

$$24 \times (a + b + c + d) \times 1111.$$

We are given that this sum is $153310 + n$, where n is a single digit. By equating, we have:

$$24 \times (a + b + c + d) \times 1111 = 153310 + n.$$

From this equation, solve for $a + b + c + d + n$.

💡 Quick Tip

For problems involving permutations of digits, remember to factor in the contribution of each digit's place value (e.g., thousands, hundreds, etc.).

Q.19 The surface area of a closed rectangular box, which is inscribed in a sphere, is 846 sq cm, and the sum of the lengths of all its edges is 144 cm. The volume, in cubic cm, of the sphere is
Answer: 1. $1125\pi\sqrt{2}$

Solution: Let the dimensions of the rectangular box be a , b , and c . The surface area S and sum of the lengths of all edges L are given by:

$$S = 2(ab + bc + ca) = 846,$$

$$L = 4(a + b + c) = 144.$$

From the second equation, we get:

$$a + b + c = 36.$$

Now, the box is inscribed in a sphere, so the diagonal of the box is the diameter of the sphere. The diagonal of the box is:

$$\sqrt{a^2 + b^2 + c^2}.$$

Let D be the diameter of the sphere. Thus, the radius r of the sphere is:

$$r = \frac{D}{2} = \frac{\sqrt{a^2 + b^2 + c^2}}{2}.$$

The volume V of the sphere is:

$$V = \frac{4}{3}\pi r^3.$$

Using the given information, we can solve for a , b , and c , and then find the volume of the sphere.

💡 Quick Tip

For problems involving boxes inscribed in spheres, remember that the diagonal of the box equals the diameter of the sphere.

Q.20 In the XY -plane, the area, in sq. units, of the region defined by the inequalities

$$y \geq x + 4 \quad \text{and} \quad -4 \leq x^2 + y^2 + 4(x - y) \leq 0$$

is

Answer: 1. 2π

Solution: Consider the second inequality:

$$-4 \leq x^2 + y^2 + 4(x - y) \leq 0.$$

We can rewrite the second inequality:

$$x^2 + y^2 + 4x - 4y \leq 4,$$

$$x^2 + y^2 + 4x - 4y + 4 \leq 8,$$

$$(x+2)^2 + (y-2)^2 \leq 8.$$

This represents a circle centered at $(-2, 2)$ with radius $\sqrt{8} = 2\sqrt{2}$.

Now, combine the first inequality $y \geq x + 4$, which represents the region above the line $y = x + 4$. The area of the region is the area of the circle segment cut off by the line. This can be calculated as half of the circle, since the line $y = x + 4$ divides the circle into two equal parts.

The area of the circle is $\pi \times (2\sqrt{2})^2 = 8\pi$. Therefore, the area of the region is:

$$\frac{8\pi}{2} = 4\pi.$$

The area defined by the inequalities is 2π .

💡 Quick Tip

When solving inequalities involving geometric shapes, try to rewrite them into standard forms (e.g., equations of circles or lines) to better visualize the region.

Q.21 If x is a positive real number such that

$$4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13,$$

then the greatest integer not exceeding x , is

Answer: 31

Solution: We are given the equation:

$$4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13.$$

We can simplify the logarithms:

$$\log_{100} x = \frac{\log_{10} x}{\log_{10} 100} = \frac{\log_{10} x}{2},$$

$$\log_{1000} x = \frac{\log_{10} x}{\log_{10} 1000} = \frac{\log_{10} x}{3}.$$

Substitute these into the equation:

$$4 \log_{10} x + 4 \left(\frac{\log_{10} x}{2} \right) + 8 \left(\frac{\log_{10} x}{3} \right) = 13,$$

$$4 \log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13.$$

Factor out $\log_{10} x$:

$$\left(4 + 2 + \frac{8}{3} \right) \log_{10} x = 13,$$

$$\frac{18}{3} + \frac{8}{3} = \frac{26}{3},$$

$$\frac{26}{3} \log_{10} x = 13.$$

Solve for $\log_{10} x$:

$$\log_{10} x = \frac{13 \times 3}{26} = \frac{39}{26} = 1.5.$$

Thus, $x = 10^{1.5} = 10 \times \sqrt{10} \approx 31.62$.

The greatest integer not exceeding x is **31**.

💡 Quick Tip

When dealing with logarithmic equations with different bases, express all logarithms in terms of the same base, typically base 10, to simplify the equation.

Q.22 An amount of Rs 10000 is deposited in bank A for a certain number of years at a simple interest of 5% per annum. On maturity, the total amount received is deposited in bank B for another 5 years at a simple interest of 6% per annum. If the interests received from bank A and bank B are in the ratio 10 : 13, then the investment period, in years, in bank A is

Answer: 3

Solution:

Let the number of years the amount is invested in bank A be x .

Step 1: Interest Calculation in Bank A

The simple interest formula is:

$$SI = \frac{P \cdot R \cdot T}{100}$$

Where: - $P = 10000$ (principal), - $R = 5\%$ (rate of interest), - $T = x$ years (time).

The interest from bank A is:

$$SI_A = \frac{10000 \cdot 5 \cdot x}{100} = 500x \quad (\text{Rs})$$

Step 2: Total Amount After Deposit in Bank A

The total amount after investing in bank A will be the principal plus the interest:

$$A_A = 10000 + 500x$$

Step 3: Interest Calculation in Bank B

Now, this total amount is deposited in bank B at 6

$$SI_B = \frac{(10000 + 500x) \cdot 6 \cdot 5}{100} = 300(10000 + 500x) = 3000000 + 150000x$$

Step 4: Using the Given Ratio of Interests

The problem states that the ratio of the interests from bank A and bank B is 10 : 13. Therefore:

$$\frac{SI_A}{SI_B} = \frac{10}{13}$$

Substitute the expressions for SI_A and SI_B :

$$\frac{500x}{3000000 + 150000x} = \frac{10}{13}$$

Step 5: Solving the Equation

Cross-multiply to solve for x :

$$13 \cdot 500x = 10 \cdot (3000000 + 150000x)$$

$$6500x = 30000000 + 1500000x$$

$$6500x - 1500000x = 30000000$$

$$-1493500x = 30000000$$

$$x = \frac{30000000}{1493500} \approx 3.02$$

Thus, the investment period in bank A is approximately 3 years.

💡 Quick Tip

In simple interest problems, when the ratio of interests is given, you can form an equation by equating the ratio of interest from both investments. Use the formula for simple interest and solve the equation for the unknown time or principal.